

Topic 4: Sustaining Ecosystems

Life on Earth is supported by global ecosystems and the link between human wellbeing and ecosystem wellbeing is vital. This topic seeks to explore the distribution and characteristics of the Earth's ecological wonders. Learners investigate the two contrasting ecosystems of tropical rainforests and

polar environments, exploring physical cycles and processes that make these ecosystems distinctive, the threats posed to their existence and how humans are attempting to manage them for a more sustainable future.

4.1. Why are natural ecosystems important?		Scale
a. What are ecosystems?	Understand the concept of an ecosystem as being the interdependence of climate, soil, water, plants and animals.	
	Outline the global distribution of polar regions, coral reefs, grasslands, temperate forests, tropical forests and hot deserts.	G
	Overview of the climate, flora and fauna within these ecosystems.	G
4.2. Why should tropical rainforests matter to us?		
a. What biodiversity exists in tropical rainforests?	- The distinctive characteristics of a tropical rainforest ecosystem, including the climate, nutrient cycle, soil profile and water cycle. - The interdependence of climate, soil, water, plants, animals and human activity in tropical rainforests.	R, L
b. Why are tropical rainforests being 'exploited' and how can this be managed sustainably?	Explore the value of tropical rainforests through the study of their goods and services.	G, R, N, L
	Human impacts in the tropical rainforest from activities such as logging, mineral extraction, agriculture and tourism.	R, L
	A case study to illustrate attempts to sustainably manage an area of tropical rainforest, such as ecotourism, community programmes, biosphere reserves and sustainable forestry, at a local or regional scale.	R, L
4.3. Is there more to polar environments than ice?		
a. What is it like in Antarctica and the Arctic?	Outline the distinctive characteristics of Antarctica and the Arctic, including climate, features of the land and sea, flora and fauna.	R, L
	The interdependence of climate, soil, water, plants, animals and human activity in either the Antarctic or the Arctic polar region.	R, L
	Explore a range of impacts of human activity on either the Antarctic or the Arctic ecosystems, such as scientific research, indigenous people, tourism, fishing, whaling and mineral exploitation.	R, L
b. How are humans seeking a sustainable solution for polar environments?	A case study to examine one small-scale example of sustainable management in either the Antarctic or the Arctic such as sustainable tourism, conservation and whaling. A case study to examine one global example of sustainable management in either the Antarctic or the Arctic by investigating global actions such as Earth Summits or the Antarctic Treaty.	G, R, L